Long Data

Improving access to and use of the city's archival records

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Long data

Big data means many records but typically over a short span

Municipal records often run continuously for decades → *long data*

Long data can illustrate trends, establish precedents, enable comparisons, and uncover context

Long data also poses challenges around accessibility, standards, comparability

Holding	**Accessible?**	**Amount**
Digital files with text	Yes	Small
Scanned pages	No	Large
Paper, film, audio	No	Unknown

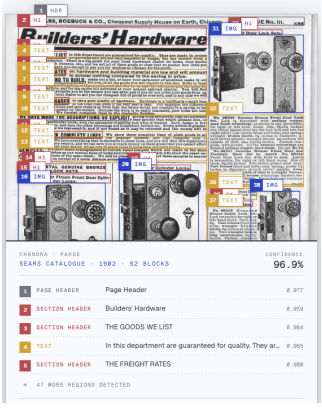
Samuel Arbesman, "Stop Hyping Big Data and Start Paying Attention to 'Long Data'," *Wired*, January 29, 2013

Document understanding

Document understanding is a research field and software frameworks that turn documents into data

- **Layout analysis** finds the headings, paragraphs, tables, and figures on a page, and the order to read them in
- **Optical character recognition (OCR)** reads printed or handwritten text from a picture of a page
- **Table recognition** rebuilds a table's rows, columns, and cells
- **Field extraction** pulls named items like dates or totals out of forms

Contemporary AI models have rapidly improving ability to convert complex PDFs into more accessible text, markdown, HTML formats



From `https://www.datalab.to`

Yihao Ding, *et al.* 2026. A Survey on MLLM-based Visually Rich Document Understanding: Methods, Challenges, and Emerging Trends. In *ACL 2026*, pp. 13319–13340.

Data liberation and rescue

Data liberation moves information that is already public out of the formats that bury it (scans, PDFs, portals) and into open and accessible data formats

Data rescue is a grassroots interdisciplinary movement to preserve public datasets from decay and censorship

- **2014.** The Sunlight Foundation runs a PDF Liberation Hackathon to build shared tools for pulling tables out of government PDFs
- **2017 on.** The Public Utility Data Liberation project turns federal energy filings into an open and documented database
- **2020.** SCOTUS rules in *Georgia v. Public.Resource.Org* that the state's annotated code cannot be copyrighted
- **2025 on.** Researchers improvise technical capacity to archive data before it is deleted by the Trump administration

Tom Lee, "PDF Liberation: Why It Matters and How You Can Help," Sunlight Foundation, January 24, 2014; *Georgia v. Public.Resource.Org, Inc.*, 590 U.S. 255 (2020)

Data liberation for the City of Boulder

- **Hotline answers.** Aiding staff in research to answer questions from council and other departments
- **Budget history.** Converting PDF tables into accessible data formats
- **The law on a date.** Which version of an ordinance was in force when a permit was issued
- **Records requests.** A document made searchable once is ready for every later request that asks for it
- **Accessibility.** Every scanned page gains text that a screen reader can speak
- **Research and reporting.** Residents, reporters, and the university work from the same corpus as staff

Two data liberation sprints

I have already piloted several data liberation sprints using local government archives

City budget books, 2002–2027

City budgets

34 PDFs and 10,843 pages from city records.

6 volumes needed to be OCR'd

450 values traced to source documents and reconciled

CDE enrollment yearbooks, 1986–2025

Pupil membership

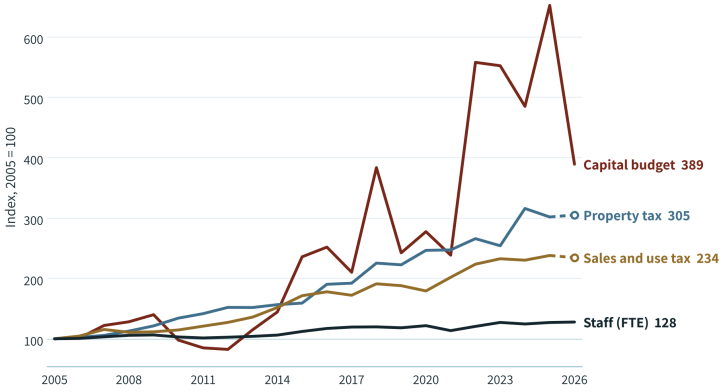
Scanned tables of enrollment by grade and school district

OCR and table extraction using Datalab

1,269 values extracted and audited for BVSD and SVVSD

Agent pipelines to identify page ranges, orchestrate OCR, extract tables, audit margins, record gaps, compare across volumes

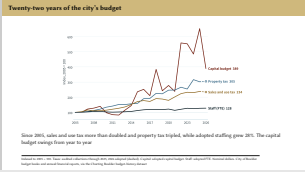
Twenty-two years of the city's budget



Since 2005, sales and use tax more than doubled and property tax tripled, while adopted staffing grew 28%. The capital budget swings from year to year

Indexed to 2005 = 100. Taxes: audited collections through 2025, 2026 adopted (dashed). Capital: adopted capital budget. Staff: adopted FTE. Nominal dollars. City of Boulder budget books and annual financial reports, via the Charting Boulder budget-history dataset

Twenty-two years of the city's budget



Taskforce prompt: what do we know today. This is the budget sprint's output. 2005 is the first year with all four series; the capital line is the adopted capital budget, a plan, not actual spending, and its dollars are nominal.

Teaching with city's records portal

Week 09 of my Web Data Science course uses Finance Department's year-end revenue reports:
`https://cuinfoscience.github.io/Web-Data-Science-Book/ch-09-pdfs.html`

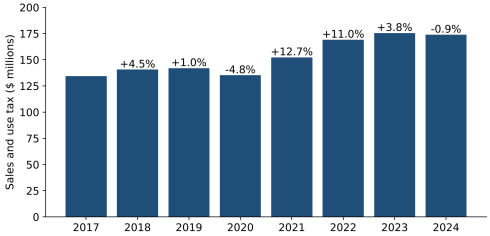
Access city's Laserfiche archives, download report PDFs, find and extract tables, check arithmetic against reported margins and prior years

Table 1 - Sales and Use Tax

All Funds

Sources - Related to Current Economic Activity	YTD through December			
	2017	2018	$ Change	% Change
Sales Tax	$104,144,137	$106,614,788	$2,470,651	2.37%
Rec MJ Additional 3.5%*	1,324,935	1,775,066	450,131	33.97%
Business Use Tax	14,216,145	13,070,617	(1,145,528)	(8.06%)
Construction Use Tax	9,897,738	13,018,049	3,120,311	31.53%
Motor Vehicle Use Tax	4,496,393	4,853,318	356,925	7.94%
Total Sales & Use Tax based on current economic activity	**$134,079,348**	**$139,331,838**	**$5,252,490**	**3.92%**
Sources - Related to Prior Periods				
Sales Tax Audit Revenue	189,929	211,748	21,819	11.49%
Business Use Tax Audit Revenue	269,615	1,105,124	835,509	309.89%
Total Sales and Use Tax	**$134,538,892**	**$140,648,710**	**$6,109,818**	**4.54%**

* Additional sales tax on Recreational Marijuana is dedicated to the general fund



Council hotline

Council members and staff exchange questions and answers via council hotline

797 messages since March 2023: 209 questions from council and 144 answers from staff

Many of those answers point to budgets, presentations, ordinance, and reports already in the city's archives

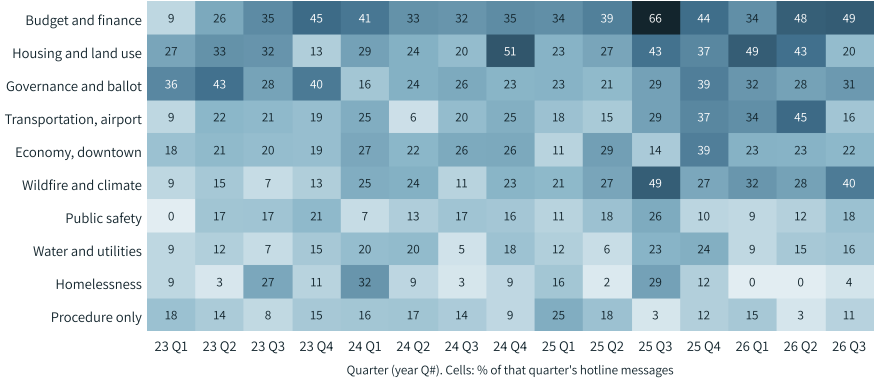
Question

How many of council's questions could be answered with staff-supervised automation using the city's own records?

City of Boulder open data, "Emails to the City Council Hotline," March 2, 2023 to October 15, 2026

Analyzing hotline: questions from council

Classify each message's subject and body based on keywords into nine policy themes



Code, keyword lists, and labels: `code/long-data-analysis.ipynb`, `code/hotline-labels-all.csv`. Messages through October 15, 2026; the chart shows complete quarters

Analyzing hotline: answers from staff

Classify each staff answer's subject and body based on keywords into eight kinds of records it draws on

Of 144 answers, the most common: plans and studies (47%), financial records (43%), program data (39%), legal and code text (38%)

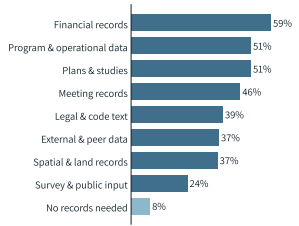
56% draw on three or more kinds of records; 14% need none

Council's questions mention financial records more often (70%) than staff answers do (43%)

A random 40 answers, coded by hand:

The answer took	Answers
A resend, a logistics note, or a transmittal	10
A lookup: it already sat in a packet, memo, ordinance, or dataset	10
Facts from the record, plus judgment	14
New analysis or a policy position	6

Kinds of records named: share of 353 questions and answers



What to liberate first

How often council needs the record? How far back it runs? How much personal information?

Start now

- Budgets, revenue, audits
- Council packets
- Staff memos
- Ordinances, resolutions
- Code versions
- Reports and plans

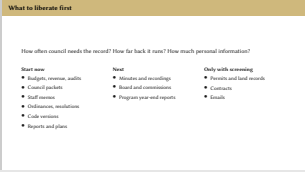
Next

- Minutes and recordings
- Board and commissions
- Program year-end reports

Only with screening

- Permits and land records
- Contracts
- Emails

What to liberate first



Taskforce prompts: implications for the city; key opportunities. Appointments are small and cheap, and they tell you who served on what, and when, which links every other record to a person's role. The appendix has the detail.

Accessibility

A scanned page is a picture. A screen reader, the software that reads a page aloud to a blind resident, finds nothing on it

Six volumes of the city's own budget books, 1,651 pages, are pictures of this kind

Federal rules under the Americans with Disabilities Act require the city's web content, posted PDFs included, to meet an accessibility standard (WCAG 2.1, level AA) by April 26, 2027. In April 2026 the Justice Department pushed that date back a year

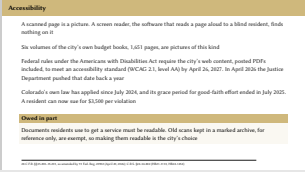
Colorado's own law has applied since July 2024, and its grace period for good-faith effort ended in July 2025. A resident can now sue for $3,500 per violation

Owed in part

Documents residents use to get a service must be readable. Old scans kept in a marked archive, for reference only, are exempt, so making them readable is the city's choice

28 C.F.R. §§35.200–35.201, as amended by 91 Fed. Reg. 20902 (April 20, 2026); C.R.S. §24-34-802 (HB21-1110, HB24-1454)

Accessibility



Taskforce prompt: key opportunities and risks. Lead with this in any conversation about money: it carries an external deadline. The archive exemption has four conditions: made before the deadline, kept only for reference or recordkeeping, unaltered, and in a clearly marked archive. An old PDF loses its exemption if residents use it to apply for or reach a service.

The Colorado Open Records Act

Colorado declares that “all public records shall be open for inspection by any person at reasonable times,” except where the law says otherwise

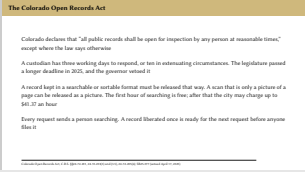
A custodian has three working days to respond, or ten in extenuating circumstances. The legislature passed a longer deadline in 2025, and the governor vetoed it

A record kept in a searchable or sortable format must be released that way. A scan that is only a picture of a page can be released as a picture. The first hour of searching is free; after that the city may charge up to $41.37 an hour

Every request sends a person searching. A record liberated once is ready for the next request before anyone files it

Colorado Open Records Act, C.R.S. §§24-72-201, 24-72-203(3) and (3.5), 24-72-205(6); SB25-077 (vetoed April 17, 2025)

The Colorado Open Records Act



Taskforce prompt: key opportunities. This is the obligation residents use most directly. Liberation doesn't change what is public under CORA; it changes how long finding it takes.

Community values

The city's Sustainability, Equity and Resilience Framework sets the goals behind its budget and comprehensive plan. Under **Responsibly Governed**, the city commits to a government that

- "Models stewardship, resilience, and sustainability of the city's financial, human, **information**, and physical assets"
- "Supports strategic decision-making with … timely and reliable data and analysis"
- "Enhances and facilitates **transparency**, accuracy, efficiency, effectiveness, and quality customer service in all city business"
- "Disaggregates data by race and ethnicity to ensure programs, policies, and financial decisions do not have a disparate impact on historically excluded community members"

The framework counts information among the city's assets, alongside money, people, and buildings. The records are the oldest part of that asset, and partners already help maintain it: the open data program works with local civic data groups, and the university law library has digitized the city charter

City of Boulder, "Sustainability, Equity and Resilience Framework," bouldercolorado.gov/sustainability-equity-and-resilience-framework

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City of Boulder, "Sustainability, Equity, and Resilience Framework," bouldercolorado.gov/sustainability-equity-and-resilience-framework

Taskforce prompts: key opportunities; what would the city need to do. These are the city's words, not mine. The fourth objective returns on the equity slide: digitizing records unevenly is a disparate impact of exactly the kind it names.

What happens next

Reading a scanned page costs under a cent today, and the price keeps falling

Finding the right document, checking what the machine read, and deciding what may be released will still take people. Retention schedules and legal review don't get cheaper with faster chips

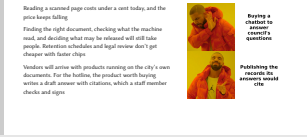
Vendors will arrive with products running on the city's own documents. For the hotline, the product worth buying writes a draft answer with citations, which a staff member checks and signs



Buying a chatbot to answer council's questions

Publishing the records its answers would cite

What happens next



Taskforce prompt: what do we think will happen. The meme is the whole recommendation in two panels.

Why I might be wrong

Demand may be low. My evidence is one coder and 40 hotline answers

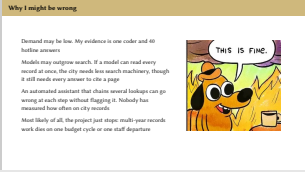
Models may outgrow search. If a model can read every record at once, the city needs less search machinery, though it still needs every answer to cite a page

An automated assistant that chains several lookups can go wrong at each step without flagging it. Nobody has measured how often on city records

Most likely of all, the project just stops: multi-year records work dies on one budget cycle or one staff departure



Why I might be wrong



Taskforce prompt: why might we be wrong. Say the demand point flatly: it is the strongest objection to my own recommendation, and the room should hear me make it first. The last point is the most likely failure, and it isn't technical.

Some records shouldn’t become data

Sort every holding into one of three kinds: **already data**, **could become data**, or **shouldn’t become data**

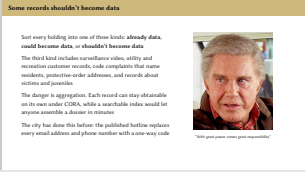
The third kind includes surveillance video, utility and recreation customer records, code complaints that name residents, protective-order addresses, and records about victims and juveniles

The danger is aggregation. Each record can stay obtainable on its own under CORA, while a searchable index would let anyone assemble a dossier in minutes

The city has done this before: the published hotline replaces every email address and phone number with a one-way code



“With great power comes great responsibility”



Taskforce prompts: key risks; mitigation. Guard against being heard as against transparency. Records law doesn’t change; what changes is whether the city builds the index. Practical obscurity, the protection of being hard to find, disappears unless someone decides to keep it.

Who bears the cost

What gets scanned gets cited. Records left on paper drop out of every search and analysis, and the least-digitized records tend to describe the least powerful residents

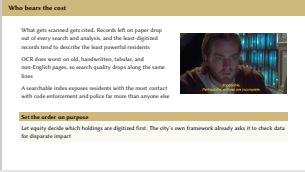
OCR does worst on old, handwritten, tabular, and non-English pages, so search quality drops along the same lines

A searchable index exposes residents with the most contact with code enforcement and police far more than anyone else



Set the order on purpose

Let equity decide which holdings are digitized first. The city's own framework already asks it to check data for disparate impact



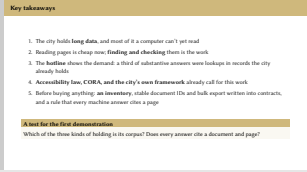
Taskforce prompt: would impacts be disproportionately felt in our community. Don't soften this into a caveat. It is a sequencing rule: it should change which holdings the city digitizes first.

Key takeaways

1. The city holds **long data**, and most of it a computer can't yet read
2. Reading pages is cheap now; **finding and checking** them is the work
3. The **hotline** shows the demand: a third of substantive answers were lookups in records the city already holds
4. **Accessibility law, CORA, and the city's own framework** already call for this work
5. Before buying anything: **an inventory**, stable document IDs and bulk export written into contracts, and a rule that every machine answer cites a page

A test for the first demonstration

Which of the three kinds of holding is its corpus? Does every answer cite a document and page?

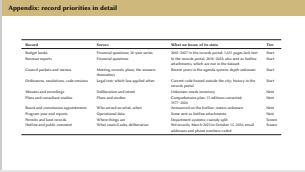


Taskforce prompt: what would the city need to do. Ten minutes in a browser and two sprints are not an audit. The finding I hold with the most confidence is the one nobody can close yet: the size of the backlog. What the city still holds after a vendor leaves is a choice it can make now, while it is cheap.

Appendix: record priorities in detail

Record	Serves	What we know of its state	Tier
Budget books	Financial questions; 26-year series	2002–2027 in the records portal; 1,651 pages lack text	Start
Revenue reports	Financial questions	In the records portal, 2018–2024; also sent as hotline attachments, which are not in the dataset	Start
Council packets and memos	Meeting records; plans; the answers themselves	Recent years in the agenda system; depth unknown	Start
Ordinances, resolutions, code versions	Legal text; which law applied when	Current code hosted outside the city; history in the records portal	Start
Minutes and recordings	Deliberation and intent	Unknown; needs inventory	Next
Plans and consultant studies	Plans and studies	Comprehensive plan: 15 editions converted, 1977–2026	Next
Board and commission appointments	Who served on what, when	Announced on the hotline; rosters unknown	Next
Program year-end reports	Operational data	Some sent as hotline attachments	Next
Permits and land records	Where things are	Department systems; custody split	Screen
Hotline and public comment	What council asks; deliberation	904 records, March 2023 to October 15, 2026; email addresses and phone numbers coded	Screen

Appendix: record priorities in detail

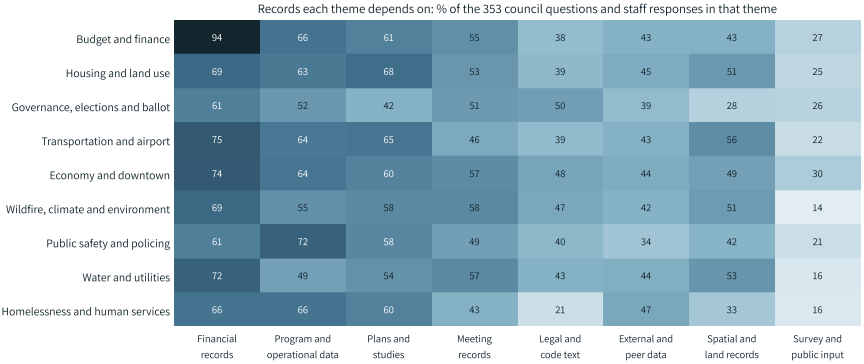


Appendix: record priorities in detail

Record	Serves	What we know of its state	Tier
Budget books	Financial questions: 10-year series	2021–2027 in the records portal; 1,421 pages back-text	Start
Revenue reports	Financial questions	In the records portal, 2018–2024; also sent as Swiftlet attachments, which are not in the records portal	Start
Council packets and memos	Meeting records, plans: the answers themselves	Recent years in the agenda system; depth unknown	Start
Ordinances, resolutions, code versions	Legal text: which law applied when	Current code hosted outside the city; history in the records portal	Start
Minutes and recordings	Deliberation and intent	Complete records; inventory	Next
Plans and consultant studies	Plans and studies	Comprehensive plan, 15 address-converted, 1977–2020	Next
Board and commission appointments	Who served on what, when	Announced on the hotline; rosters unknown	Next
Program year-end reports	Operational data	Some sent as Swiftlet attachments	Next
Permits and land records	What's changing, and where	Mixed-format; depth unknown	Soon
Hotline and public comment	What caused sales, deliberation	Voicemails, March 2023 to October 11, 2026; email addresses and phone numbers coded	Soon

Each “unknown” is a line on the inventory.

Appendix: which records each theme depends on



Financial records come up in more than 60% of questions and answers in every theme. Most answers need several kinds of record at once

Appendix: getting a report from the records portal

```
import requests, time
# PORTAL, LISTING, and DOWNLOAD hold the portal's addresses; chapter 9 gives them in full
HEADERS = -"User-Agent": "WebDataScience/1.0 (your-email@colorado.edu)"
# The folder's listing: the request the folder page itself sends (194345 is Revenue Reports)
body = -"folderId": 194345, "getNewListing": True, "start": 0, "end": 100
response = requests.post(PORTAL + LISTING, json=body, headers=HEADERS, timeout=60, verify=bundle)
for entry in response.json()["data"]["results"]:
   print(entry["entryId"], entry["name"])

time.sleep(30) # wait between requests to the city's server
response = requests.get(PORTAL + DOWNLOAD.format(docid=194397), headers=HEADERS, timeout=120, verify=bundle)
print(response.headers["Content-Type"], len(response.content), response.content[:5])
# application/pdf 723920 b'%PDF-'
```

`bundle` is a certificate file the book provides, because the city's server is set up in a way browsers forgive and Python doesn't. Chapter 9 also sends the request fields the page adds

Appendix: reading and checking the totals row

```
import pdfplumber

def rowwords(page, label):
   """Return the words to the right of label, on the line where label is."""
   found = page.search(label)[0] # where the label sits on the page, in points
   line = page.crop((0, found["top"] - 1, page.width, found["bottom"] + 1))
   return [w["text"] for w in line.extractwords() if w["x0"] ¿ found["x1"]]

for name in REPORTS:
   with pdfplumber.open(name) as pdf:
      page = findsummarypage(pdf) # the page whose text includes "This report provides"
      earlier, this, change = [tonumber(t) for t in rowwords(page, "Total Sales and Use Tax")[:3]]
   if this - earlier != change: # the report's own change column is the check
      print(f"-name: -this:, - -earlier:, is not -change:,")
# (nothing printed: every change column checks)
```

`to_number` turns "$134,538,892" or "(1,145,528)" into a whole number; accountants write a negative number in parentheses